Lecture 7

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- Finish reading Chapters 10-12 for next time.
- Begin reading Chapters 13-14.
- We will transitioning into probability.
- Quiz 3 is today at the end of lecture.
- Quiz 4 is next Thursday in lab.

Preliminary Question

Find the 35th and 80th percentiles of the following list: 4, 10, 1, 4, 1, 9, 6

Reading Summary – Chapter 10

Section 10.1: Introduction

- The regression line for y on x estimates the average value for y corresponding to each value of x.
  - Note: only works when the data follows a linear trend.
- Associated with each increase of one SD in x there is an increase of only r SDs in y on average.

Section 10.2: The Graph of Averages

- **If the data follows a linear trend,** then the regression line is a smoothed version of the graph of averages.

Section 10.3: The Regression Method for Individuals

- To estimate the average value of y for a given value of x,
  1. Convert x to standard units
  2. Multiply by r. This gives you the average value of y in standard units.
  3. Convert from standard units back to the original scale for y.
- The same procedure works for percentiles:
  - Convert to and from standard units using the normal table.
- This only works when the data follows a linear trend.

Section 10.4: The Regression Fallacy

- In virtually all test-retest situations, the bottom group on the first test will on average show some improvement on the second test, and the top group will on average fall back.
  - This is the **regression effect.**
- The **regression fallacy** is thinking the regression effect is due to something important;
  - in fact, it is just due to scatter in the data.
- Origin of the term regression: **"regression to mediocrity"**

Section 10.4: The Regression Fallacy (continued)

- The regression effect does not imply that everyone above average on the first test will do worse on the second test.
  - It only applies **on average.**
- People who do above average on the first exam still tend to do better than average on the second exam, just not by as much.
  - Similarly for those who do worse than average.

Section 10.5: There are Two Regression Lines

- The following two lines are different:
  - the regression line for y on x, which uses x to predict y, and
  - the regression line for x on y, which uses y to predict x.

Warm-up Discussion Question

- A class has a midterm and a final. On each exam, the mean is 60, and the SD is 15. A student scores at the 70th percentile on the midterm. What would be your best guess as to the student's percentile on the final?
  - a little higher than the 70th percentile,
  - a little lower than the 70th percentile, or
  - right about the 70th percentile.
- What score would be the 70th percentile on the final?

More Warm-up Discussion Questions

- What score would you predict on the final if the mean on the final is 75?
- What score would you predict on the final if the mean on the final is 50?

Reading Summary – Chapter 11

Section 11.1: Introduction

- Error = Actual - Predicted
- How big are the errors, typically?
  - Can't just take the average – it's always 0!
  - Instead: Use the Root Mean Squared (RMS) error – the square root of the average of the squared errors.
- RMS error of a regression is analogous to the SD of a list
  - RMS error indicates how far typical points are from the regression line.
  - SD indicates how far typical points are from average.

Section 11.1: Introduction (continued)

- About 68% of the points are within 1 RMS error of the regression line.
- About 95% of the points are within 2 RMS errors of the regression line.
- This is true only when the data is football shaped!

Section 11.2: Computing the RMS Error

- Method 1:
  - Compute the error for each point (need to use regression method).
  - Square all the errors.
  - Average the squared errors.
  - Take the square root of the result.
- Method 2:

$$\text{RMS error} = \sqrt{1 - r^2} \times \text{SD}(y)$$

Section 11.3: Plotting the Residuals

- **Residuals:** another word for the errors.
- Plotting the residuals can help you detect non-linear patterns in the data.

Section 11.4: Looking at Vertical Strips

- **Homoscedastic**: the SD is about the same in every vertical strip.
- **Heteroscedastic**: different vertical strips have different SDs.
- Football shaped data is homoscedastic
- For football shaped data, the SD in any vertical strip is roughly equal to the RMS error.

Section 11.5: Using the Normal Curve Inside a Vertical Strip

- Within a vertical strip...
  - We can find the average value of y by the regression method.
  - We can find the SD of y in a vertical strip by computing the RMS error.
  - We can then find percentiles and percentages using the normal approximation.
  - Only works for football-shaped data!

Final Score Interval Prediction

- Suppose that a student scored in the 70th percentile on the midterm.
- The correlation between the score on the midterm and the score on the final was 0.6
- The mean and standard deviation on the final were 75 and 15, respectively.
- We would predict that a roughly 68% chance that the student's final exam score was between _____ and _____.
- We would predict a roughly 95% chance that the student's score was between _____ and _____.
- This assumes that the distribution of scores is roughly football-shaped.

Reading Summary – Chapter 12

Section 12.1: The Regression Line

- The slope of the regression line is $r * SD(y)/SD(x)$.
- The regression line goes through the point of averages.
- Don't use regression lines from observational studies to predict the results of an intervention.

Section 12.2: The Method of Least Squares

- The regression line is the line that has the smallest RMS error among all lines.

Section 12.3: Does the Regression Make Sense?

- Consider known relationships, like between area and perimeter, before blindly applying a regression.
- Other variables frequently explain the relationship between two variables of interest.

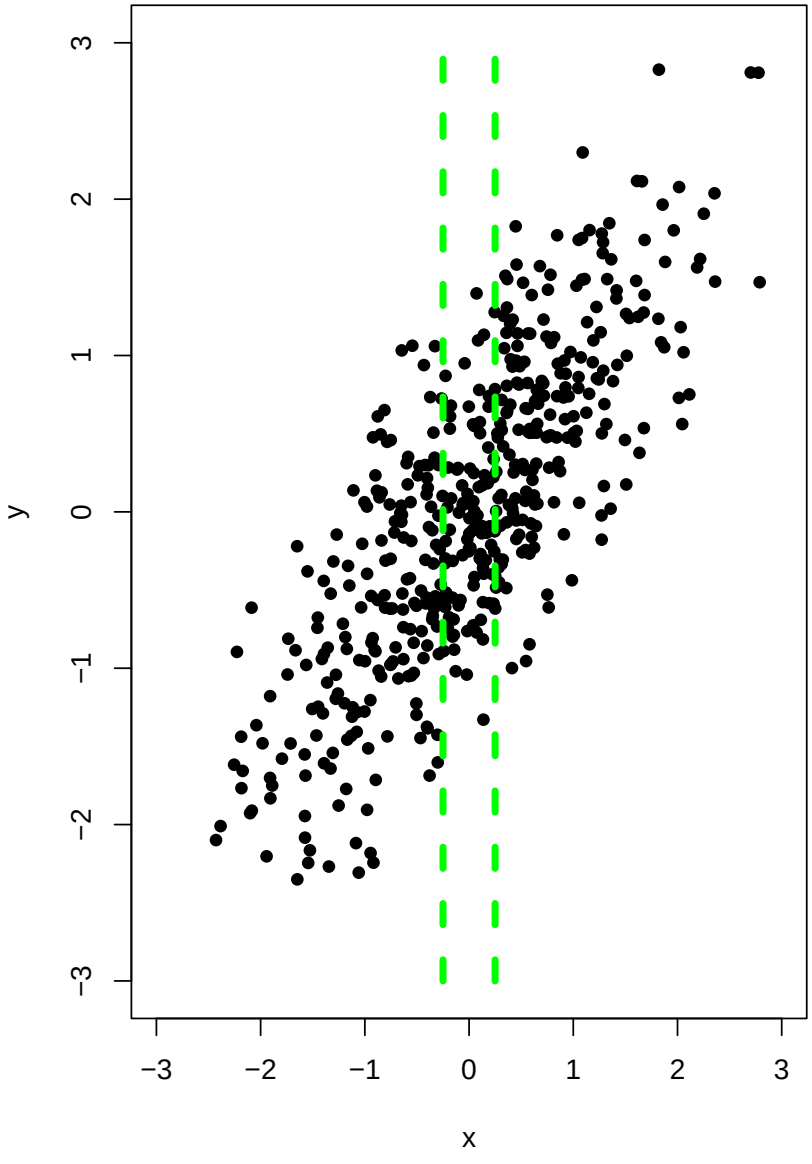
Predicting the Final Score Based on the Midterm Score

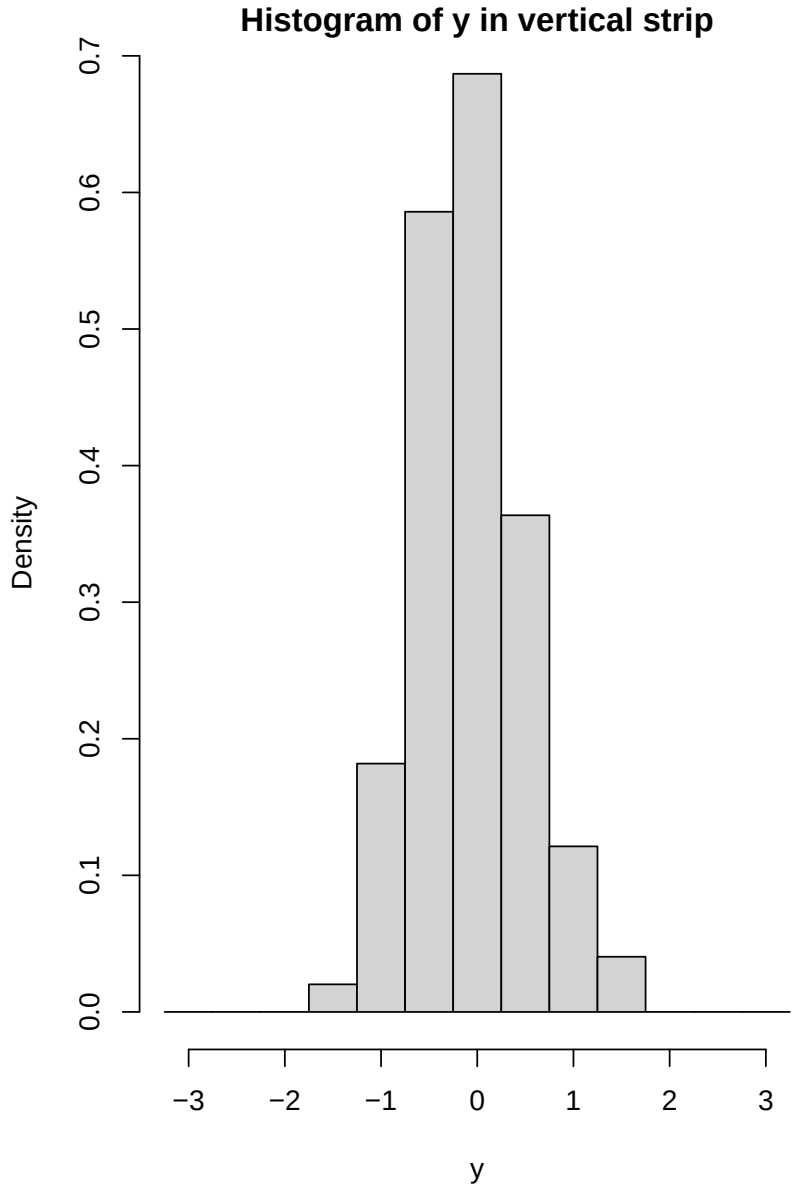
- The mean and standard deviation on the final were 75 and 15, respectively.
- The mean and standard deviation on the midterm were 60 and 15.
- The correlation between the midterm and the final scores was 0.6.
- Write an equation predicting the final score y based on the midterm score x.

Marginal and Conditional Distributions

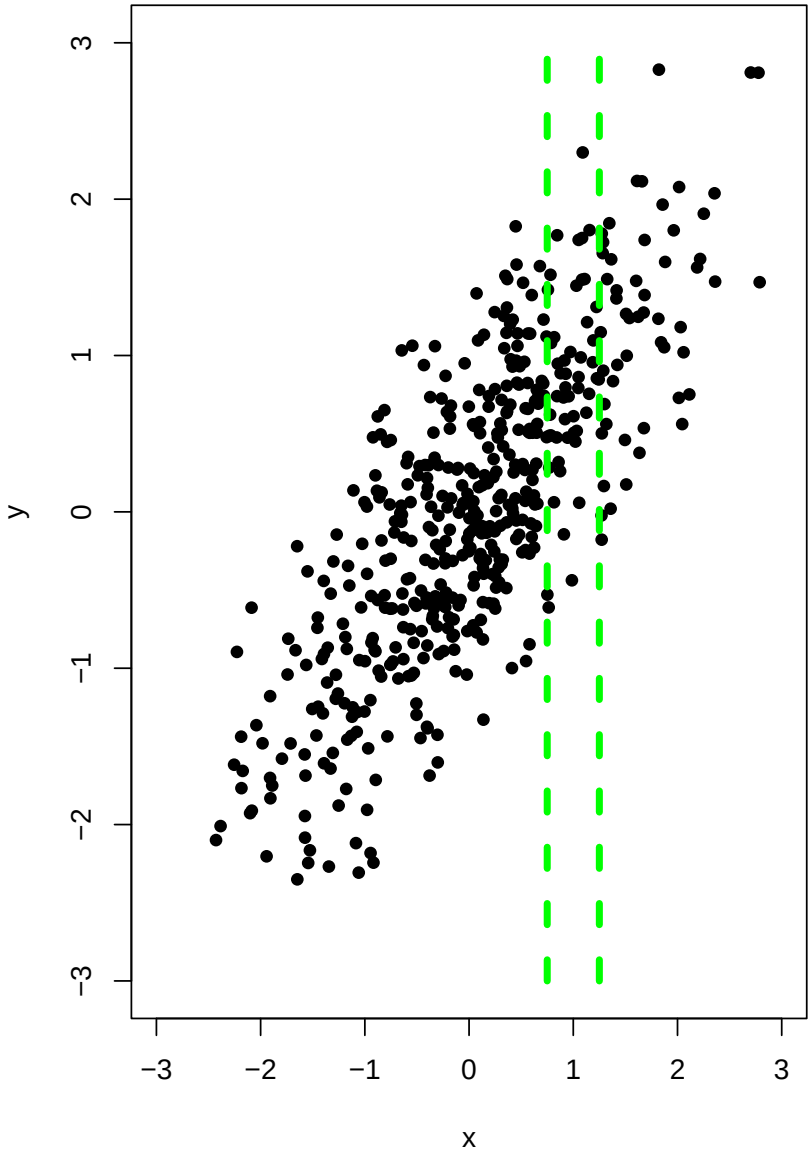
- The **conditional distribution** of y given $x \approx a$, denoted as $f(y|x \approx a)$, refers to the distribution of y in the **subset** of the data where x is approximately equal to some value a.
- In other words: the distribution of y in a vertical strip.
- The **marginal distribution** of y is the distribution of **all** of y, ignoring x.
- In other words: the distribution of y overall.

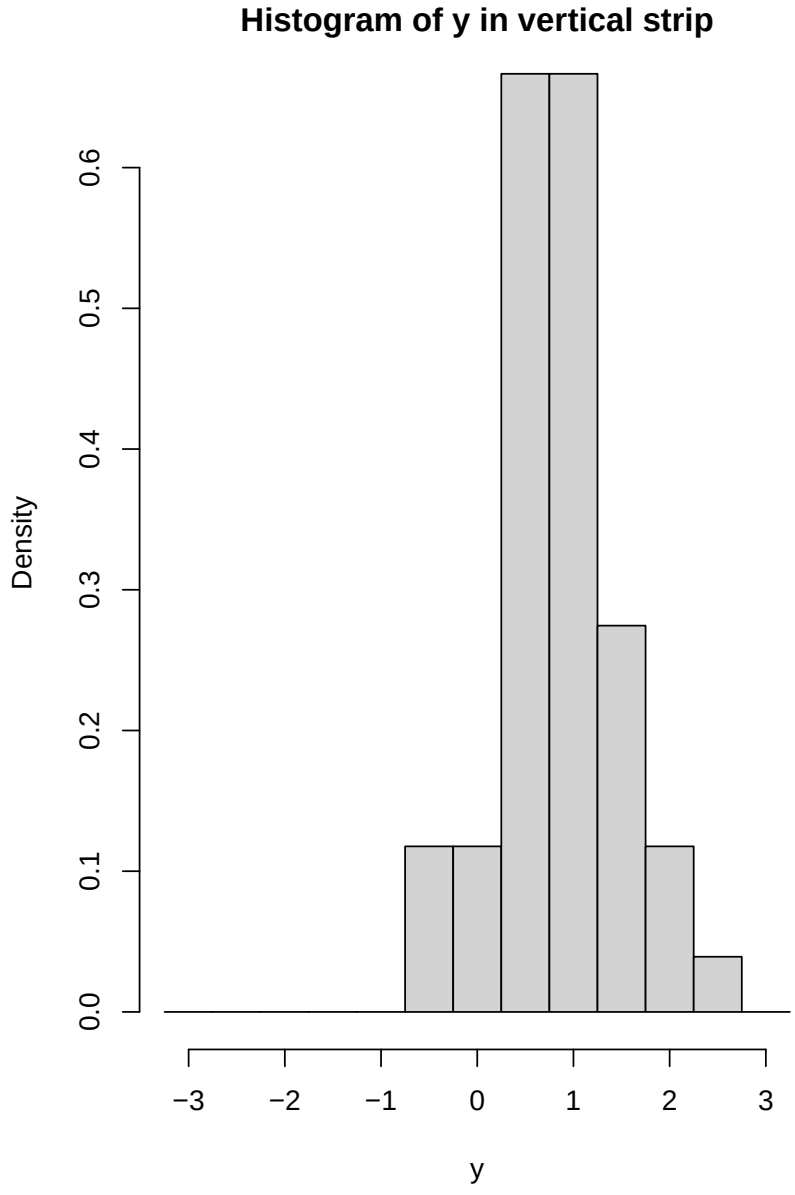
Conditional Distribution $f(y|x \approx 0)$



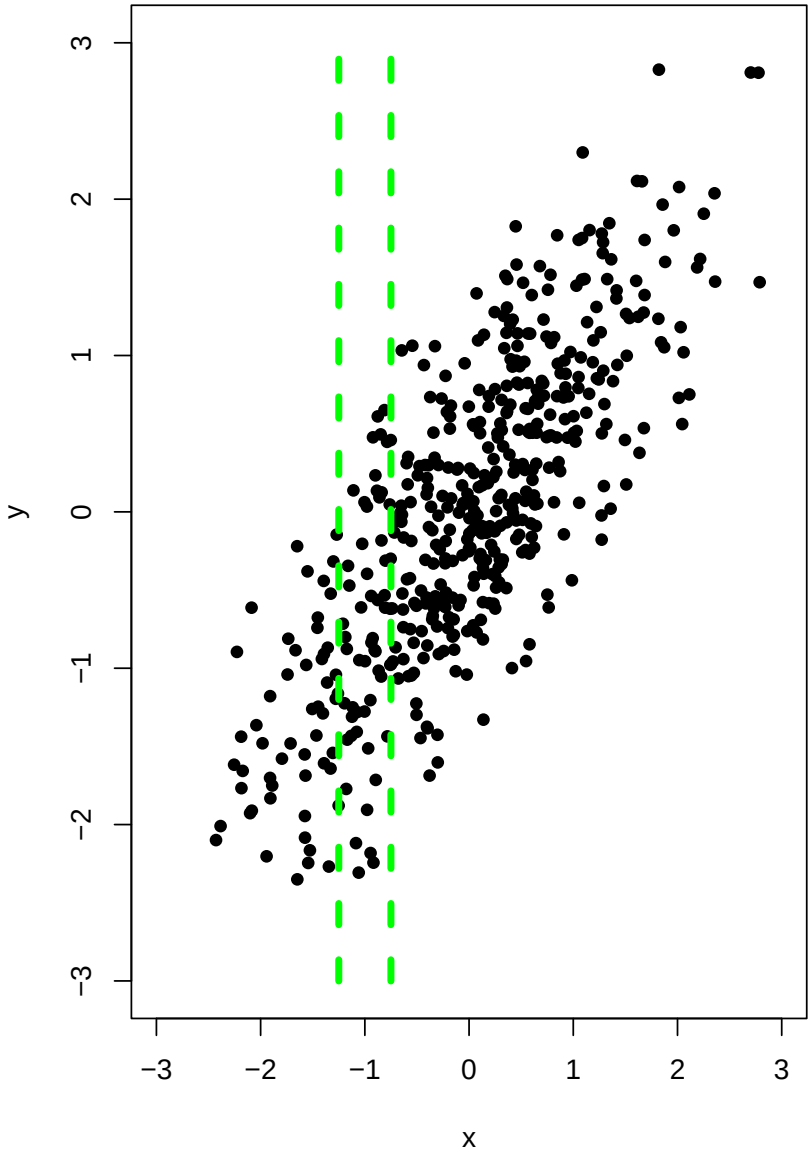


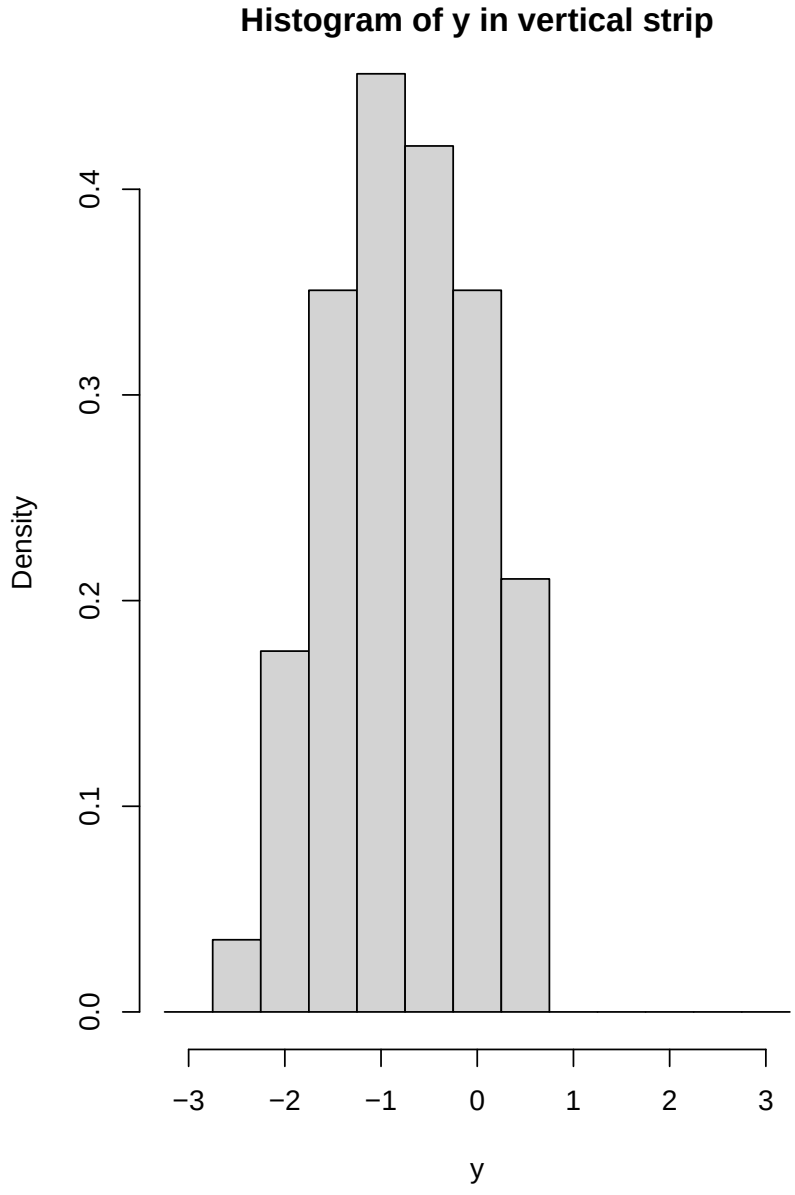
Conditional Distribution $f(y|x \approx 1)$



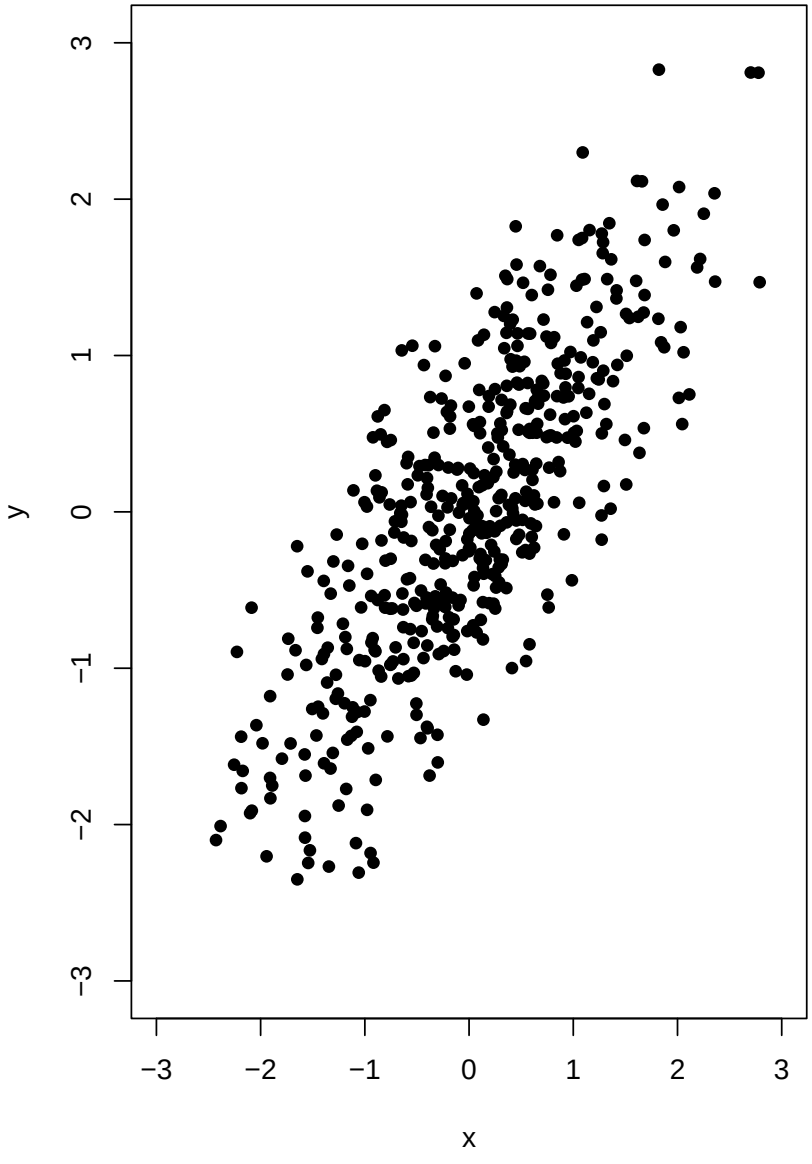


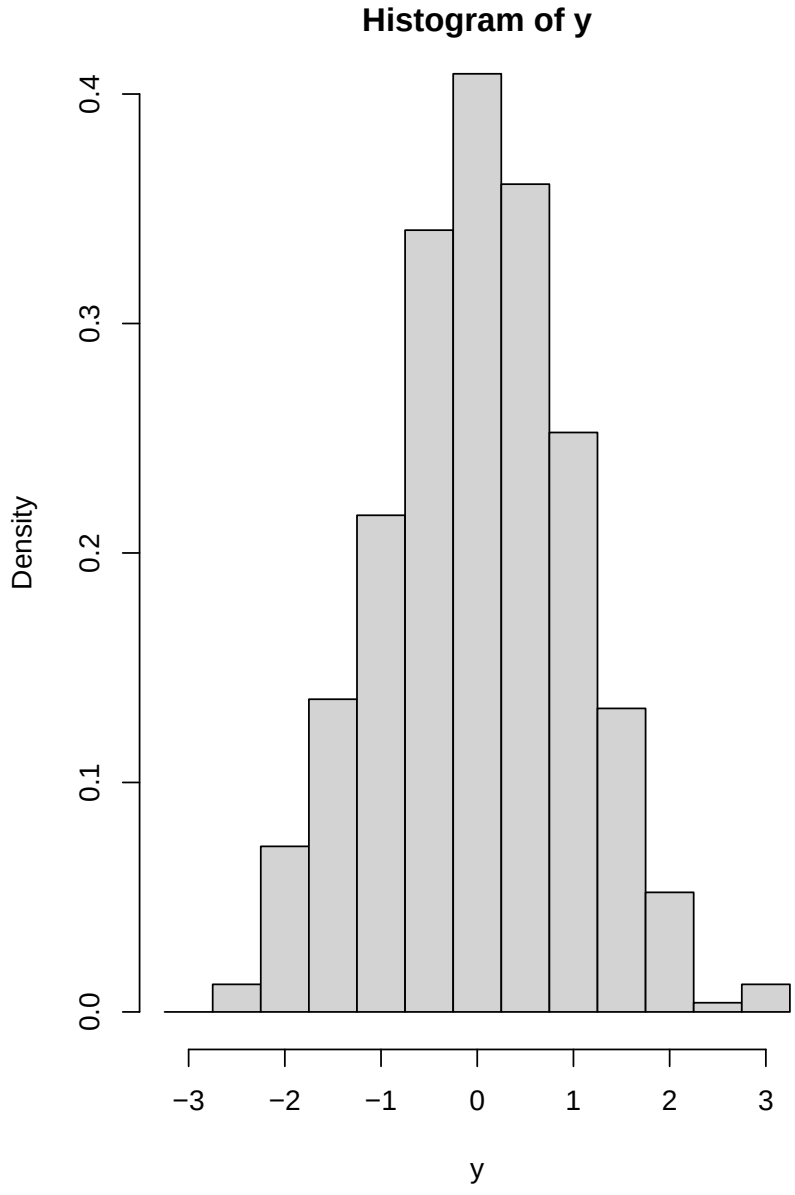
Conditional Distribution $f(y|x \approx -1)$





Marginal Distribution of y

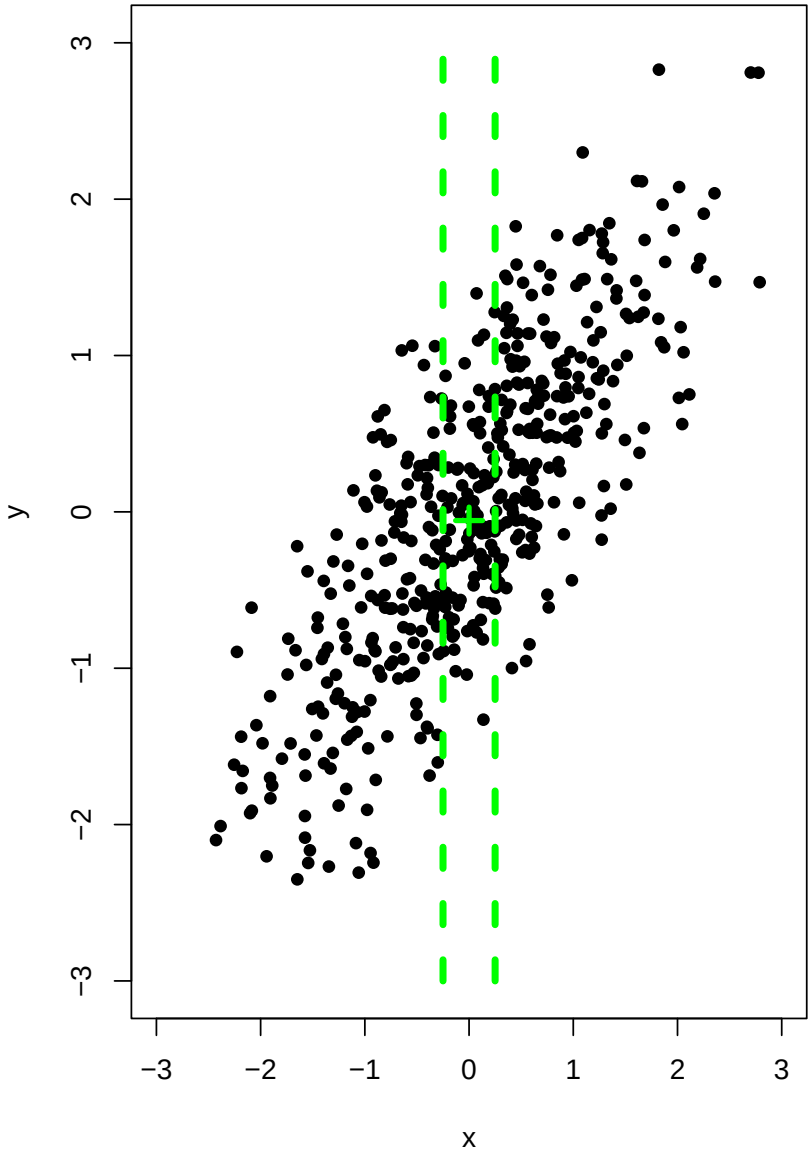


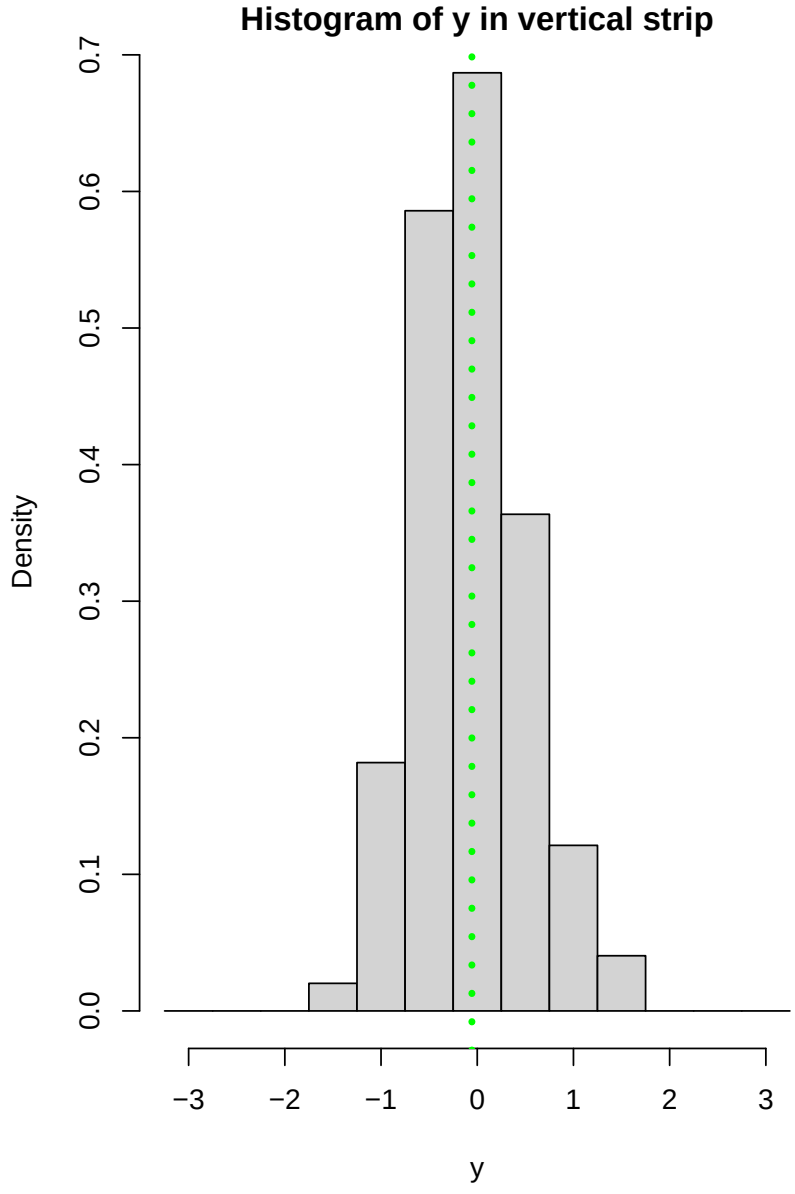


Conditional Average and Graph of Averages

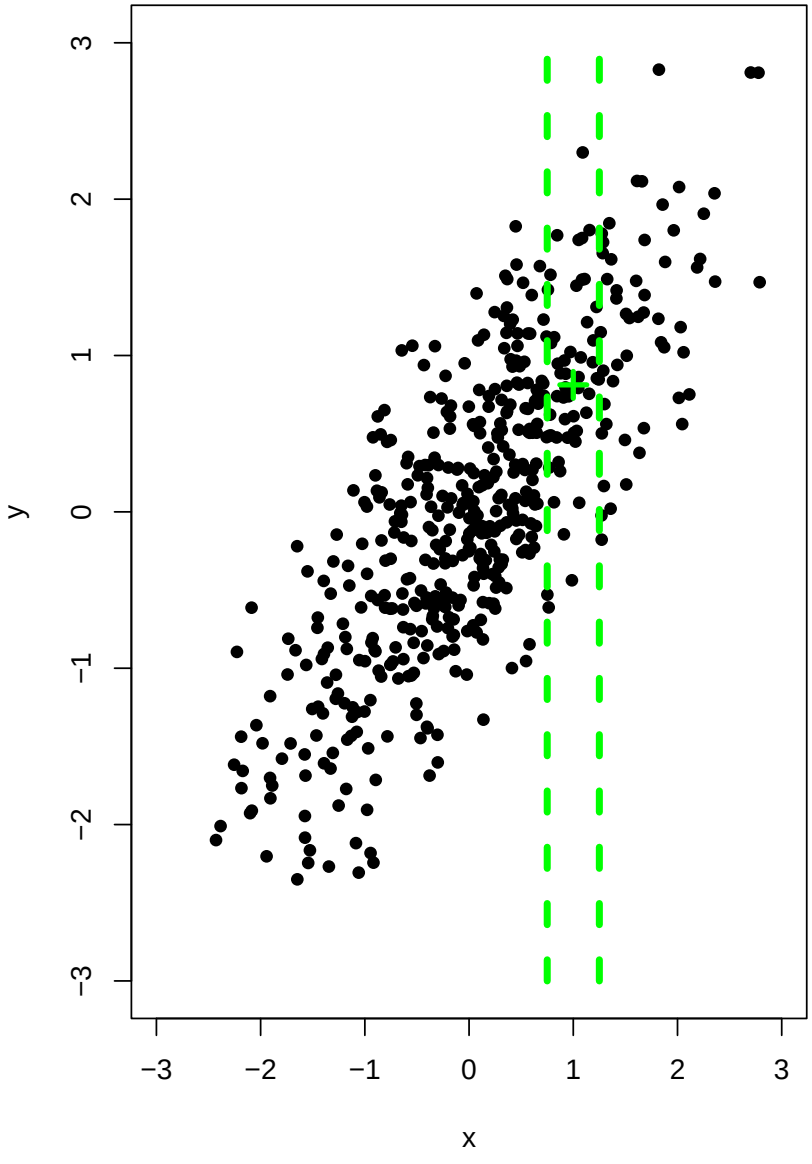
- The **conditional average** of y given $x \approx a$, denoted as $E(y|x \approx a)$, represents the average value of y within the vertical strip where x is approximately some value a.
- The **graph of averages** is the graph of $E(y|x \approx a)$ for varying values of a.

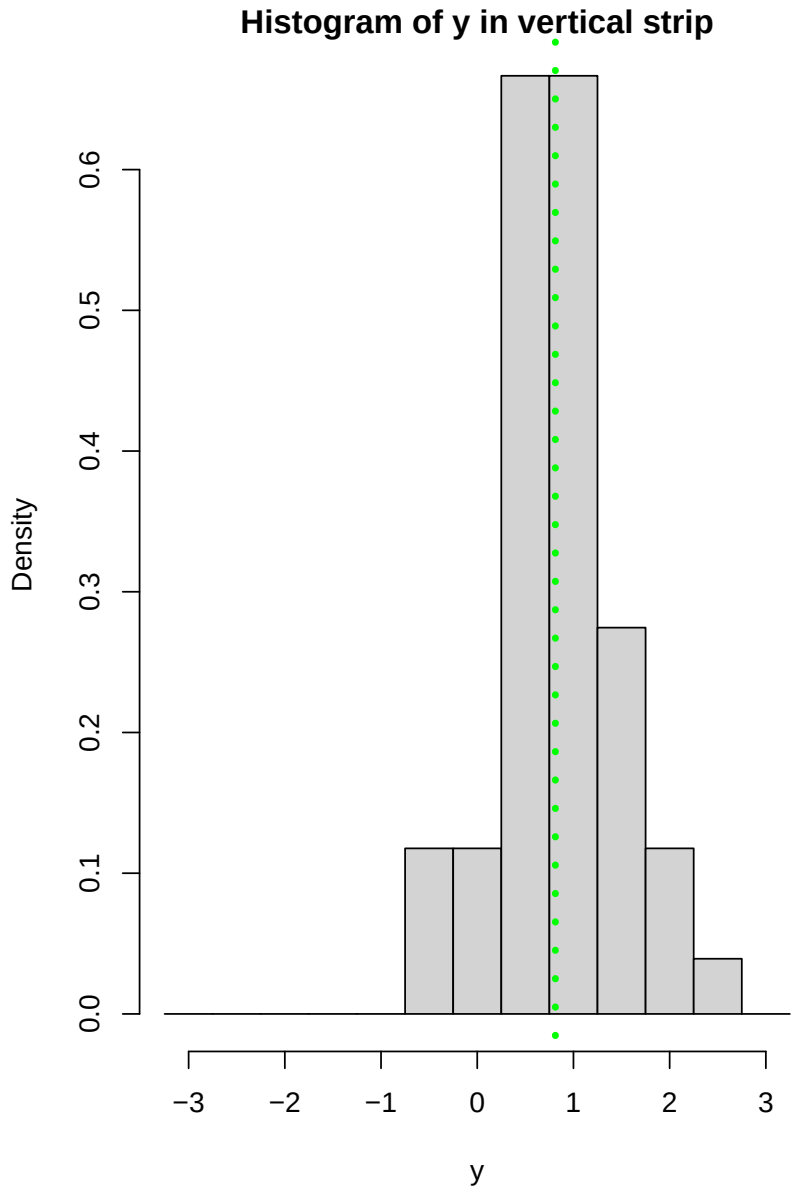
Conditional Average $E(y|x \approx 0)$



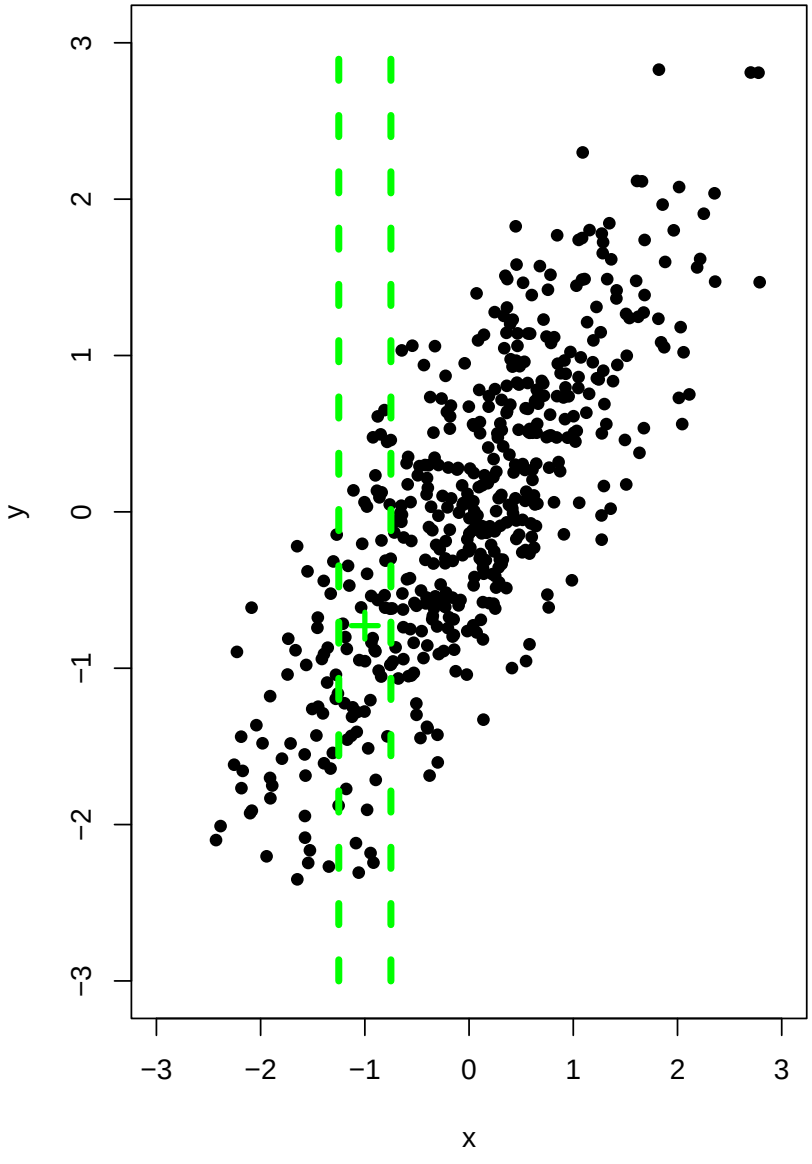


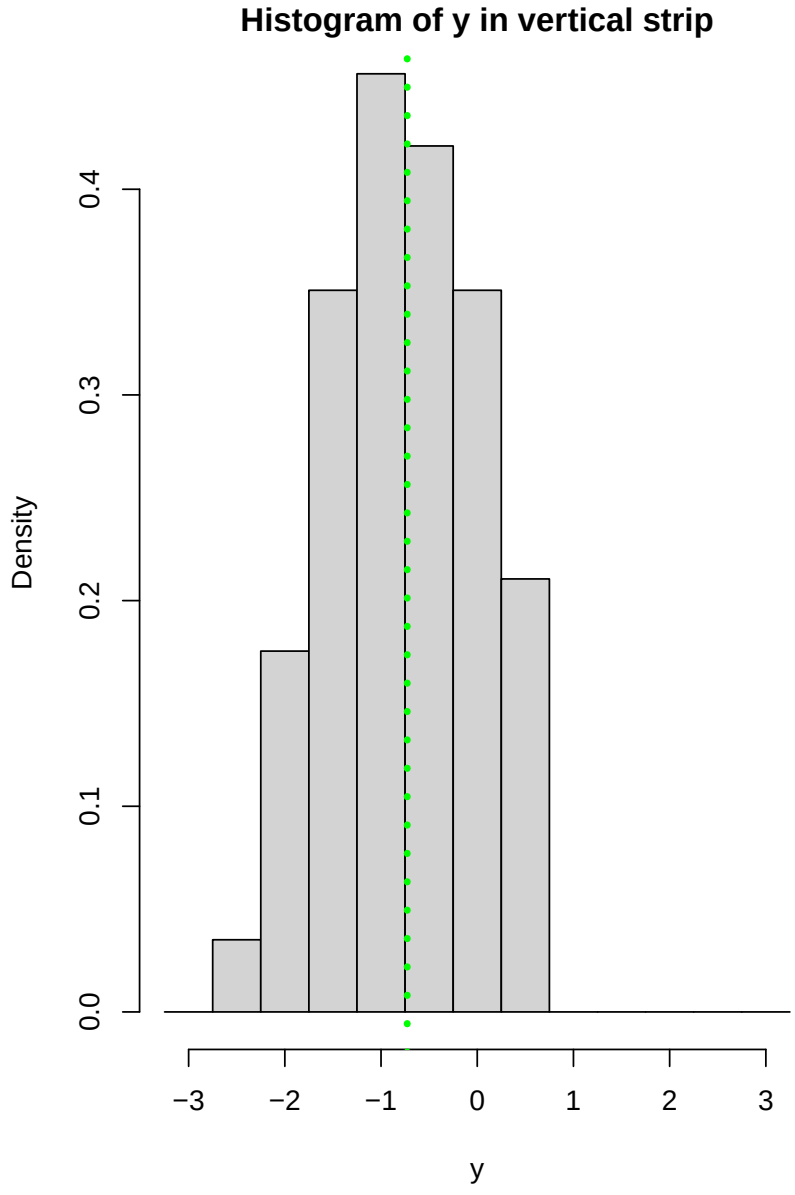
Conditional Average $E(y|x \approx 1)$



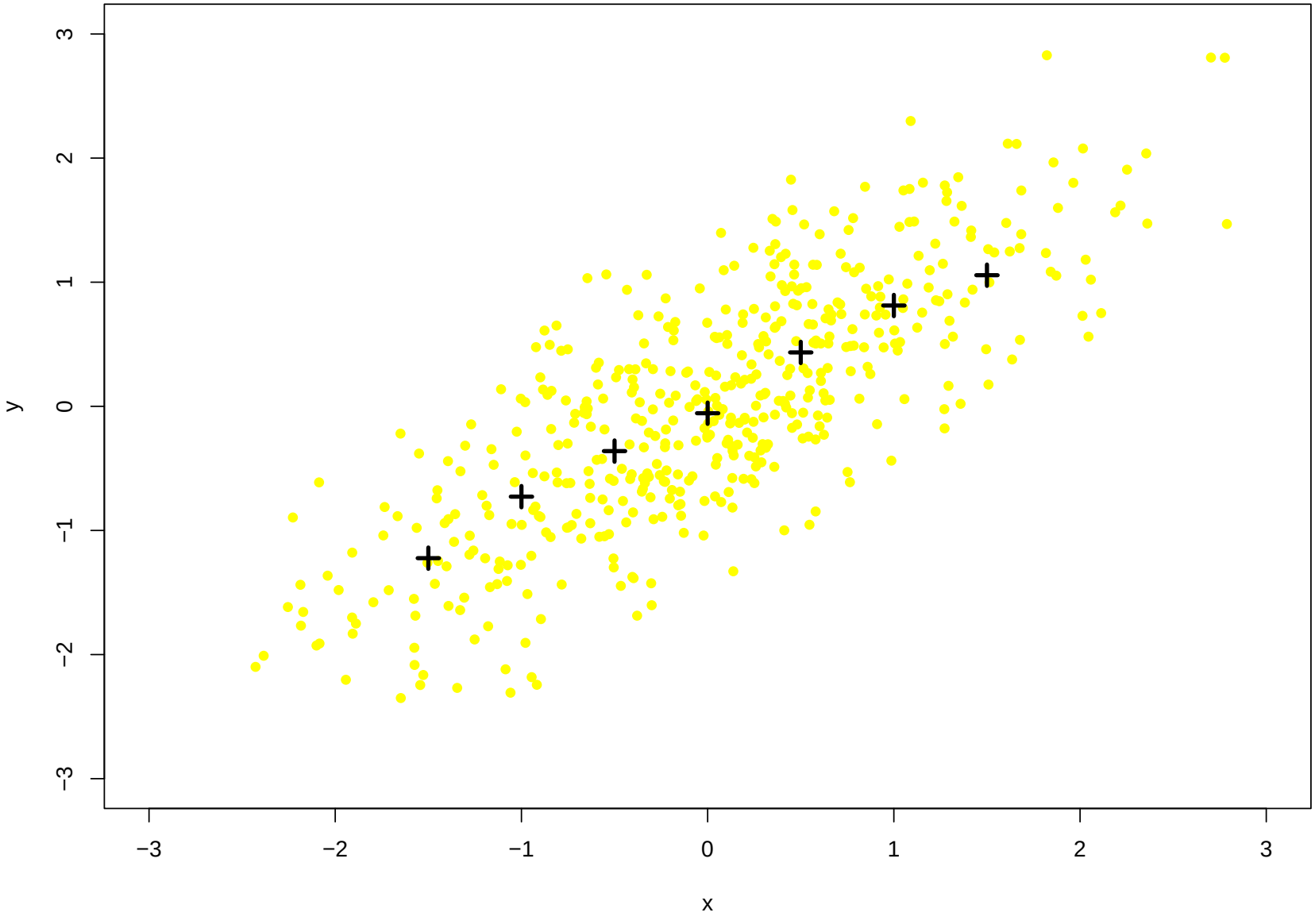


Conditional Average $E(y|x \approx -1)$



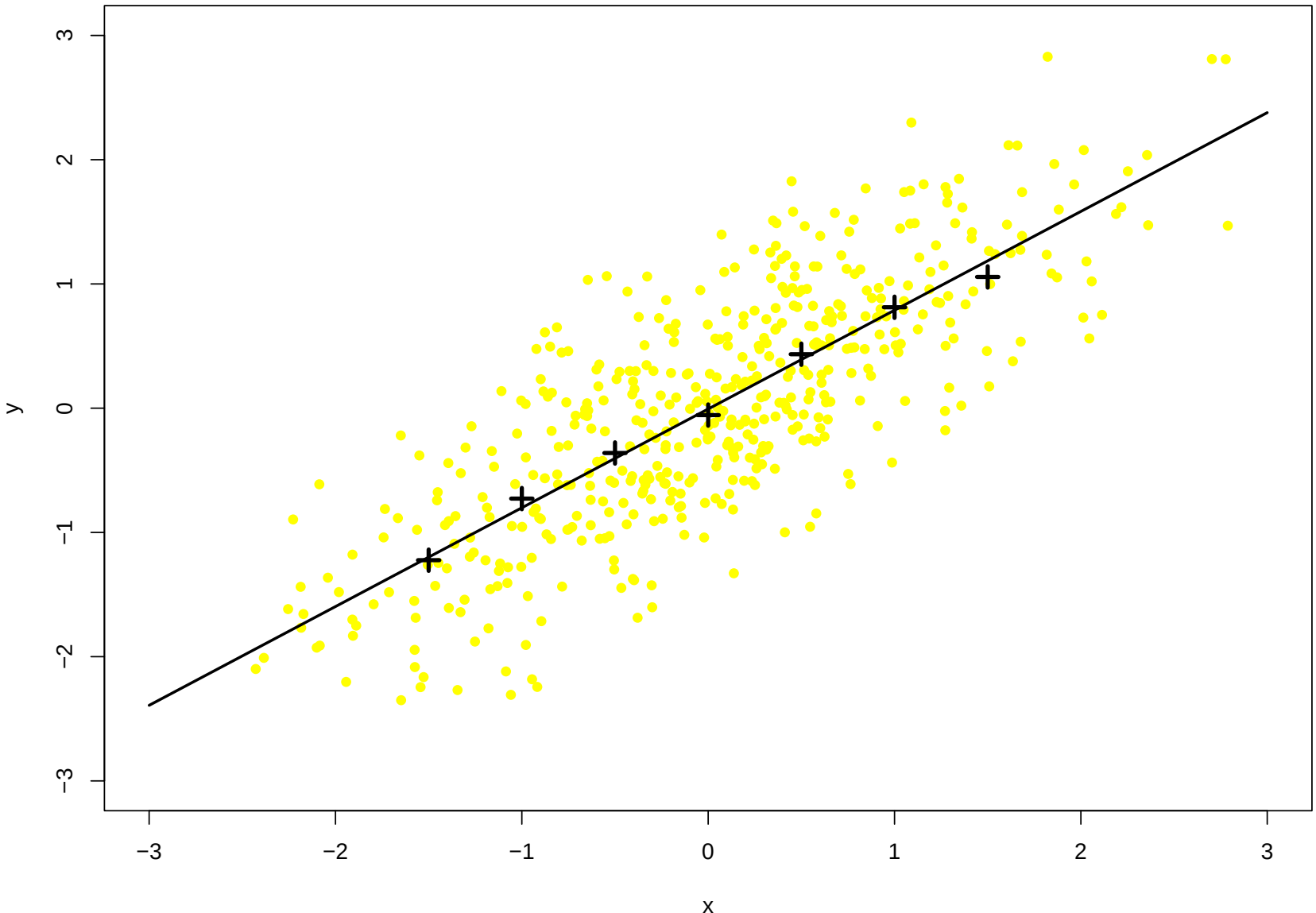


The Graph of Averages

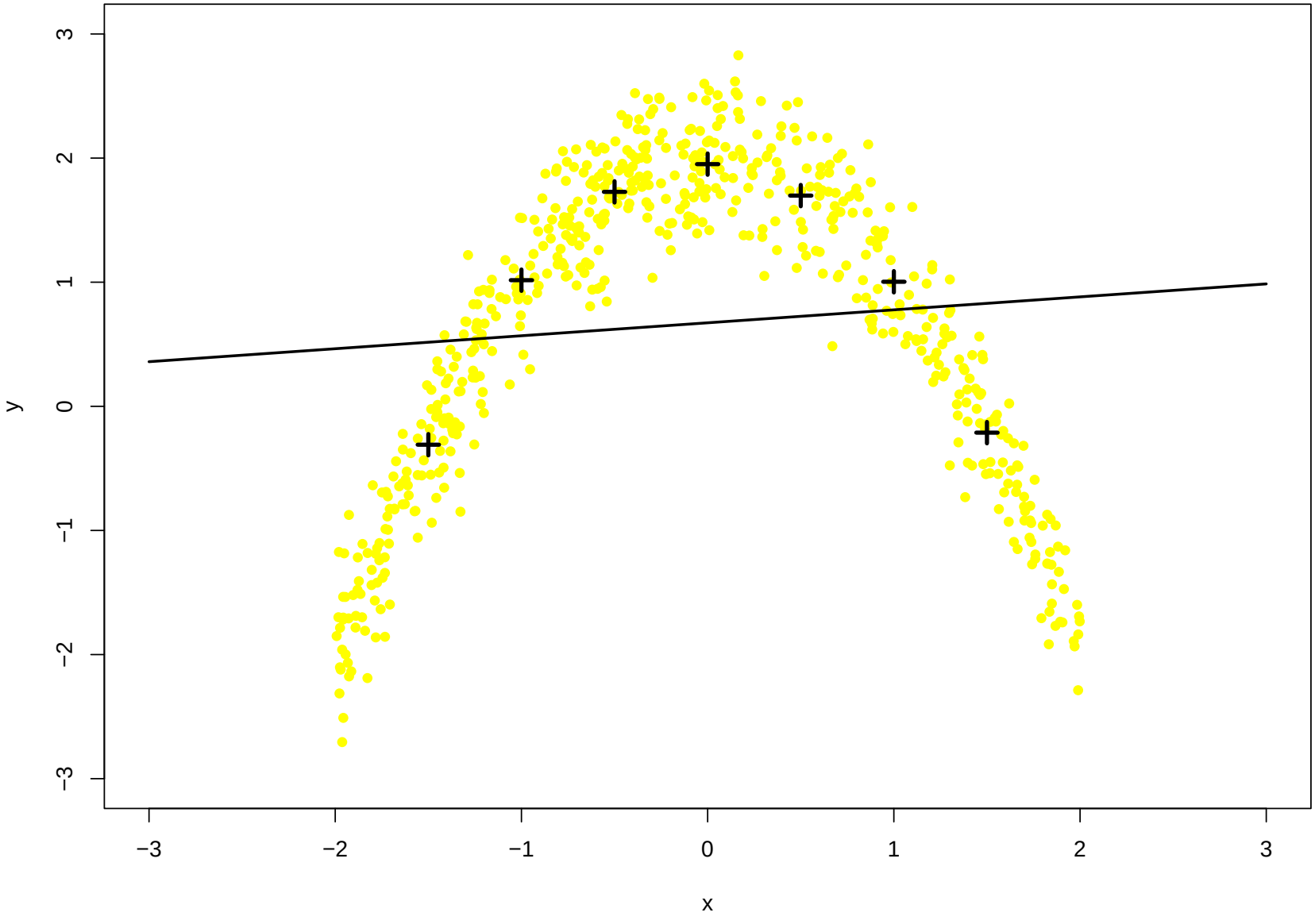


The Regression Line

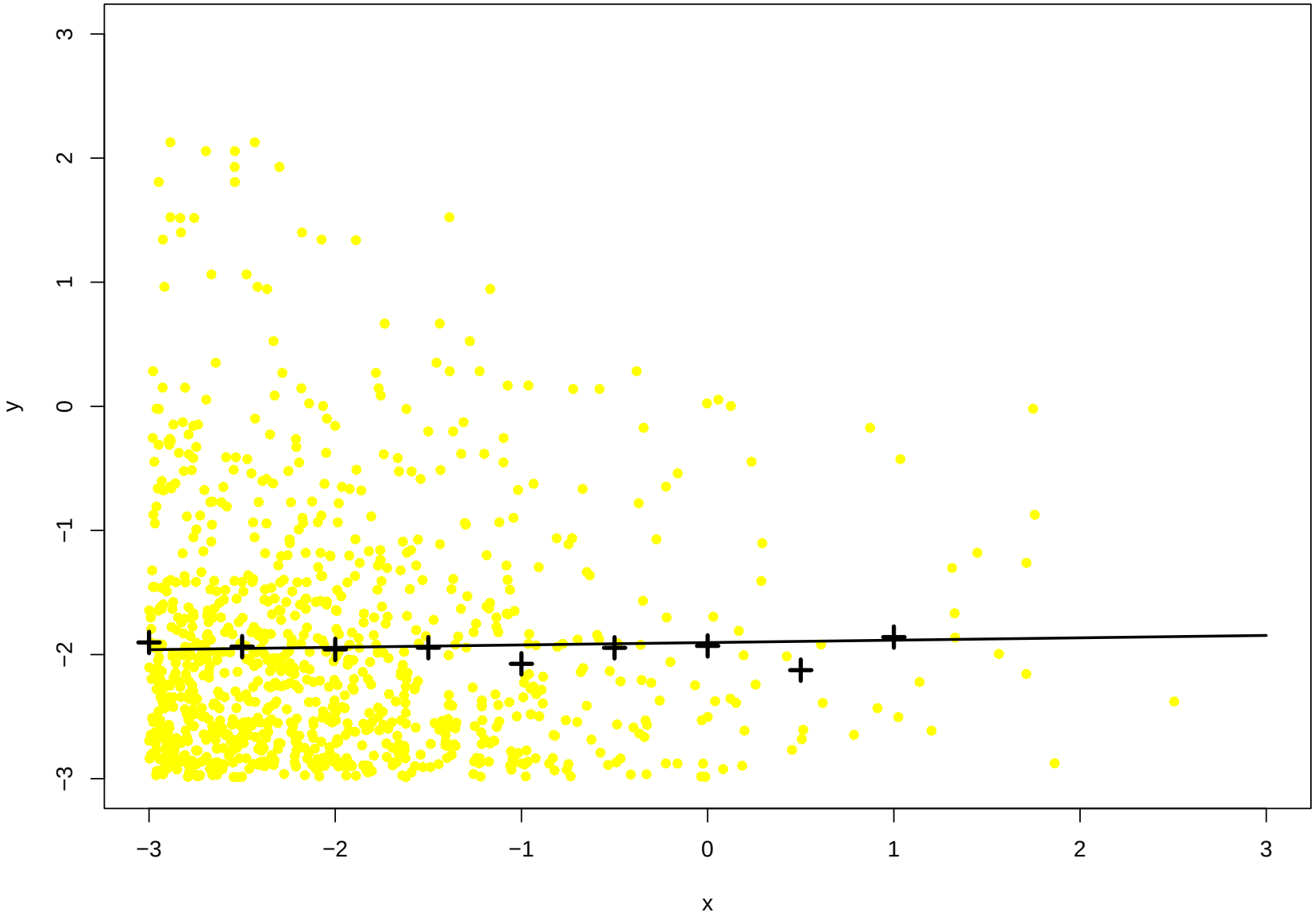
- Regression line: smoothed version of the graph of averages.
- True only when the data follow a linear trend!



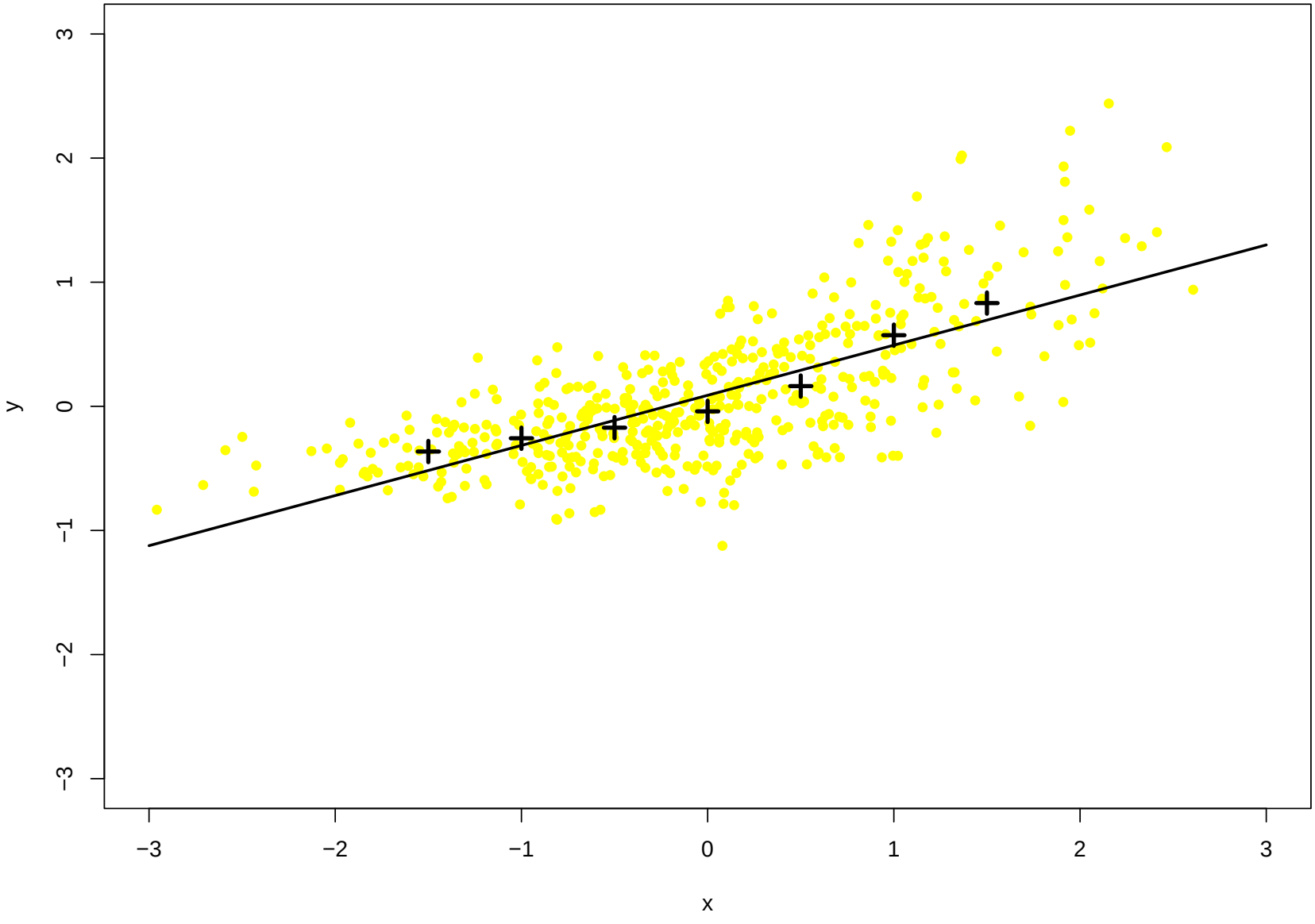
Regression Line and Non-linear Data



Regression Line and Non-linear Data Part 2



Regression Line and Non-linear Data Part 3



Where Does the Regression Line Come From?

- We know the regression line will be of the form $y = mx + b$.
- What are m and b?

Review Questions

- See Review_questions7.pdf